Marble Sort: Game Design, Physics aur Layout Spec

Source video: Marble-Sort-SnapYT_App.mp4 (30 sec, portrait 296x640, 30 fps, Level 35). Neeche ke saare numbers 296x640 reference screen ke pixels hain, video se measure kiye (±3px). AI Studio mein inhe scale karna hai: scale = actualScreenWidth / 296.

1. Game ek line mein

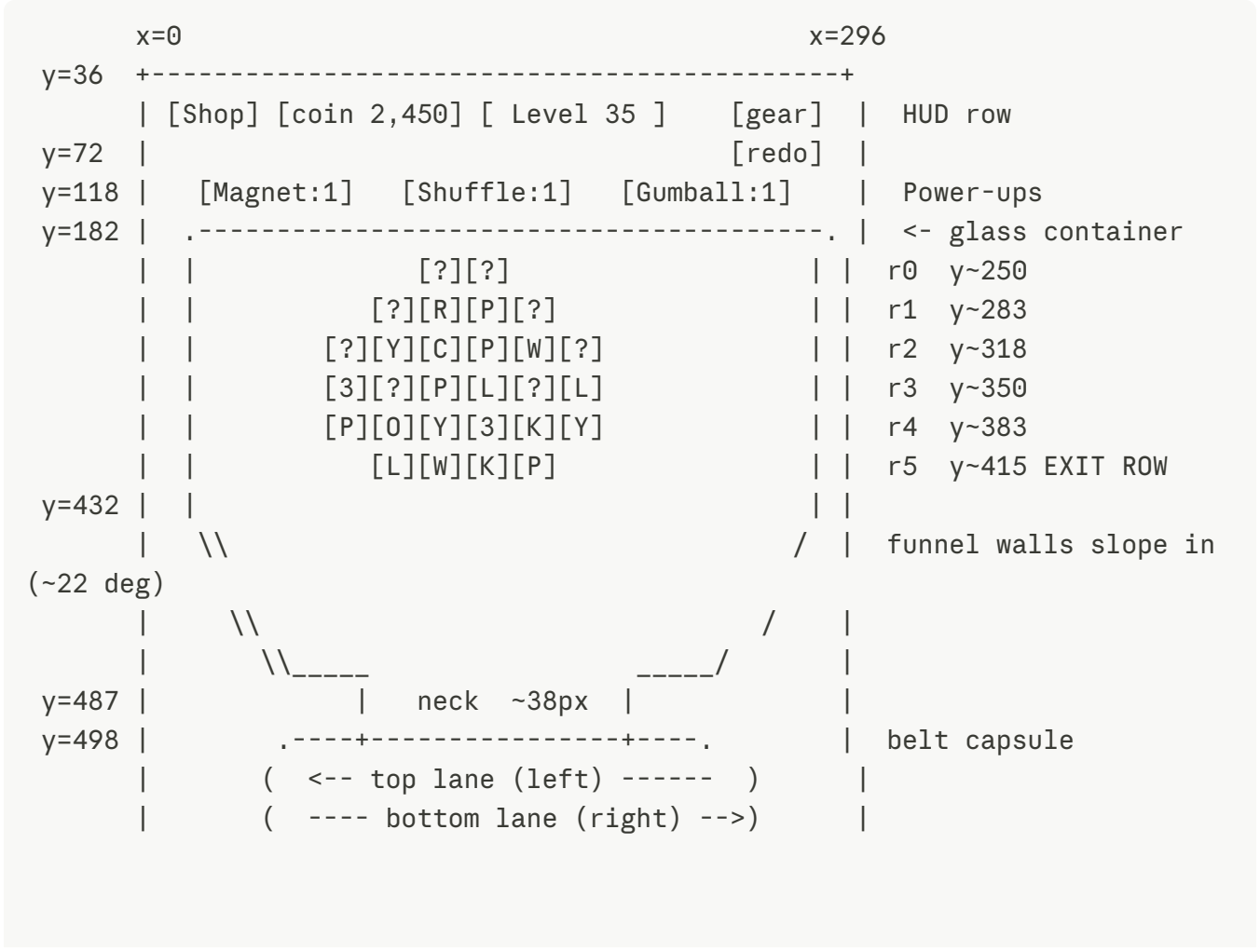
Upar glass container mein rang-birange blocks ka grid hai. Player jo block khula (exposed) ho use tap karta hai. Block se balls phat kar nikIte hain, funnel mein girte hain, tang neck se guzarte hain aur capsule shape ki conveyor belt par aate hain. Belt gol ghoomti hai. Belt ke neeche 4 columns mein bricks ki stacks hain. Belt ka ball agar kisi column ki top brick ke color se match kare to us brick ke hole mein gir jata hai. Brick ke 3 holes bhar jayen to brick pop hoke gayab, agli brick upar aa jati hai. Saari bricks khatam = level win.

2. Video mein kya kya hua (timeline)

Time	Event
0.2 s	Neeche wali row ka purple block tap hua: dark/pressed ho gaya
0.3-0.5 s	Block gayab, uski jagah se lagbhag 9-12 purple balls ek cluster mein phate
0.6-1.0 s	Balls funnel ki dhalwan par girte hain aur neck ki taraf slide karte hain
0.6-0.9 s	Purple ke upar wala pink block bounce karke 'khula' ho gaya (flat se bumpy texture)
1.0-2.0 s	Balls neck se ek stream mein belt ke top lane par girte hain
1.5 s	Pink block tap hua. Uske upar ka '?' tile 2.0 s par cyan reveal hua
3.0-3.5 s	White aur yellow blocks tap hue

Time	Event
3.5 s	Pehli purple brick ke 3 holes bhare, brick upar uthi, white stars, gayab. Neeche wali brick upar aayi
5-12 s	Belt par 4-5 colors ke balls ek saath ghoom rahe hain, sirf matching wale brick mein girte hain
17-22 s	Green balls ka bada pile funnel mein jama: belt busy hai to balls funnel mein wait karte hain (backlog)
23-24 s	Dispenser block ka counter 3 se 2 hua aur neeche khali cell mein naya cyan block aaya
24.5 s	Belt lagbhag poori bhar gayi (dono lanes)

3. Poori screen ka layout (diagram)



```
 y=542 |        '-------------------------------'   |
 y=548 |    [col0]     [col1]     [col2]     [col3]   |  brick stacks (screen
ke neeche tak)
 y=640 +-----------------------------------------+
```

HUD aur power-ups

Element	x	y	Size	Detail
Shop button	8-50	36-78	42x42	Pink/magenta rounded square, shop-awning icon, radius 10
Coin pill	64-148	36-68	84x32	Lavender-purple #9A7AE6, white bold text '2,450', gold coin icon left
Level pill	160-246	36-68	86x32	Same style, text 'Level 35'
Settings	262-291	36-66	29x30	Purple rounded square, white gear
Restart	262-291	72-104	29x30	Purple rounded square, white circular arrow
Power-up 1	52-105	118-174	53x56	Green square, radius 10, dark green 2px border, magnet icon
Power-up 2	122-175	118-174	53x56	Orange tangled-arrows icon (shuffle)
Power-up 3	193-246	118-174	53x56	Gumball machine icon
Badge	bottom-right of each		18px circle	Red circle, white border, white count '1'

4. Zones: shape, border, corner, color

Zone	Shape / position	Border	Corner	Fill
Background	full screen	none	none	flat teal #56A9B4

Zone	Shape / position	Border	Corner	Fill
Glass container (upar)	x 8-280, y 182-432	dark teal #427E8E, 5px + 2px light inner rim #BFE8F0	top corners radius ~32, sides straight	#93D8E4 (halka lighter top)
Grid board	stepped arch/pyramid silhouette behind blocks, centered x=147	3px light blue-grey outline	radius 8 at steps	#B9CCE0
Funnel	container ke neeche, walls (12,438) se (128,487) aur (284,438) se (166,487)	same dark teal 5px	neck lips radius 6	#C7D2EB
Neck	x 128-166, y 487-497	dark teal		opening ~38px wide
Belt housing	x 20-272, y 490-640+	dark teal-blue, 3px	top corners radius 22	#C7D2EB
Belt capsule	x 28-267, y 498-542	light rim #EEF0FA 3px	full pill (radius 22)	track dark grey #6E6E7E
Belt divider bar	x 55-240, y 518-523	none	radius 3	#EFEDFB
Belt slots	dark pill depressions 14x10 along both lanes (shows capacity)	none		#5F5F6E
Brick columns	4 stacks, centers x = 65, 120, 175, 230	none	brick radius 6	per-color

Ball/block colors (approx, eyedropper se refine karna): purple #7B3FE4, pink #FF8FD6, ice-white #E6F4FA, yellow #FFD21F, lime #A8F028, cyan #27C4F4, orange #FF8A1F, red #F23B5C. Locked tile '?': grey-blue #5E7488 with white '?'. Dispenser: blue #6B96DC with white number and dark slot line under it.

5. Block grid

Grid 6 columns x 6 rows. Cell pitch: 35px horizontal, 33px vertical. Cell size 30x30, radius 7, gap ~4. Shape mask (. = deewar / koi cell nahi):

```
          c0   c1   c2   c3   c4   c5
 x center  60   95  130  165  200  235
 r0 y250   .    .    ?    ?    .    .
 r1 y283   .    ?    R    P    ?    .
 r2 y318   ?    Y    C    P    W    ?
 r3 y350   D3   ?    P    L    ?    L
 r4 y383   P    O    Y    D3   K    Y
 r5 y415   .    L    W    K    P    .   <- EXIT ROW (neeche funnel khula)
```

Legend: ? = locked, D3 = dispenser (counter 3), R red, P purple, Y yellow, C cyan, W white/ice, L lime, O orange, K pink. Yeh Level 35 ka starting layout hai, apne level data ke liye template ki tarah use karo.

Block types

Type	Kaisa dikhta hai	Behaviour
Normal block	Rounded square, solid color, neeche 3px darker bevel	Flat = abhi tap nahi ho sakta. Bumpy = tappable
Locked '?'	Grey-blue tile, white '?'	Tap nahi hota. Jab uska koi orthogonal neighbor cell khali ho jaye to flip/scale animation se hidden color reveal hota hai, phir normal block ban jata hai
Dispenser (N)	Blue block, number N, neeche slot	Static, tap nahi hota. Jab uske seedhe neeche wala cell khali ho aur N > 0 to us cell mein naya block drop karta hai (0.25 s slide) aur N ghat jata hai (3 se 2 video mein dikha)

Exposed (tappable) rule

Block tappable tab hai jab uske 4 orthogonal neighbors mein se kam az kam ek cell khali ho (mask ke andar ka khali cell), ya wo exit row (r5) mein ho jahan neeche funnel khula hai. Mask ke bahar ki deewar khali nahi gini jati. Video se verify hua: r4c0 purple 30 sec tak flat raha kyunke uske charon taraf deewar/block/dispenser the. Khula hone par block ka 'pop' hota hai: scale 1.0 -> 1.15 -> 1.0 (0.25 s) aur texture flat se bumpy (3x3 pop-it dots) ho jata hai.

Important: video mein blocks par gravity NAHI hai. Block apni jagah rehta hai jab tak tap na ho. Sirf dispenser naye block bharta hai.

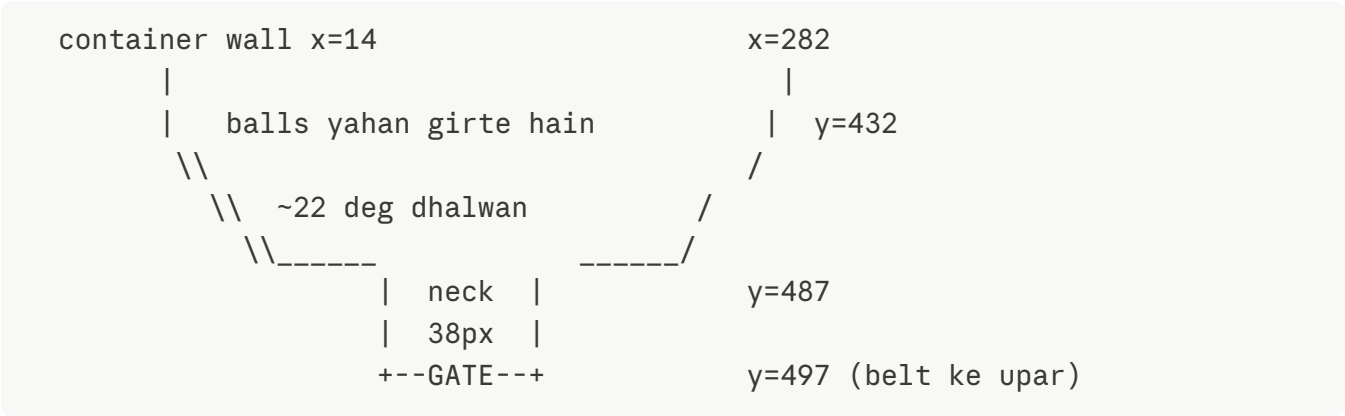
6. Tap se balls niklne tak (pop physics)

10 fps zoom se nikala hua sequence:

```
t=0.0   tap on exposed block
t=0.0-0.1  press: scaleY 0.8, scaleX 1.1, color 15% darker
t=0.1-0.3  block gayab (scale 0 + fade), N balls spawn block ke center par
t=0.3-0.5  cluster phailta hai (~40px circle), balls girte hain
t=0.5-0.9  balls khali cells se gujarkar funnel ki dhalwan par
t=0.6-0.9  neighbor block ka pop-up bounce (exposed ho gaya)
```

- Balls per block: lagbhag 9-12 dikhe. Tunable constant BALLS_PER_BLOCK. Recommended 9 (3 bricks ke barabar, neeche level-solvability rule dekho).
- Spawn: block center ke aas paas 8px radius mein random, har ball ka delay 0-60 ms (burst feel).
- Initial velocity: random direction, speed 60-140 px/s, thoda neeche ki taraf bias.
- Ball radius funnel mein ~7px (diameter ~14).
- Balls blocks se collide NAHI karte. Video mein balls grid ke upar se pass hote dikhte hain (khali cells ke saath saath blocks ke aage). Balls sirf container walls, funnel walls aur ek dusre se collide karte hain.
- Render order: grid blocks peeche, balls aage.

7. Funnel physics



Parameter	Value (296 px reference)
Gravity	~1000 px/s^2 neeche
Ball radius	7 px
Restitution (bounce)	0.25-0.35
Friction (ball-wall)	0.02 (dhalwan par aasani se slide ho)
Air drag	0.01
Ball-ball collision	ON, thode soft (balls ka pile banta hai)
Funnel wall slope	~22 degrees, wall thickness 5 px
Neck width	~38 px (2-3 balls ek saath)
Gate	Neck ke neeche ek invisible sensor. Belt ke feed point par free slot ho tabhi ball gate se guzar sakta hai, warna ball funnel mein wait karta hai

Video mein 17-22 s par green balls ka bada pile funnel ke neeche jama raha aur ek ek karke belt par jaata raha. Yeh gate ka seedha saboot hai.

8. Conveyor belt

Belt physics se nahi, path-following se chalti hai (balls fixed spacing mein chalte hain).

```
     feed point (x=147, neck ke neeche)
              |
              v
  top lane  y=508   <------------------------ (left ki taraf)
 ( left U-turn )                          ( right U-turn )
  bottom lane y=532 ------------------------>  (right ki taraf)
       ^ col0     ^ col1     ^ col2     ^ col3   (x=65,120,175,230)
```

Direction: counter-clockwise. Top lane mein balls right se left, left end par neeche, bottom lane mein left se right, right end par upar wapas top lane.

Parameter	Value
Path	Stadium: straight segments 186 px (x=57 se x=243) + 2 half circles radius 12 (centers (57,520), (243,520))
Path length	2x186 + 2x pi x12 = ~447 px
Lane y	top 508, bottom 532
Ball radius on belt	6 px (funnel ke 7 px se chhota: belt par aate waqt scale 1 -> 0.85)
Spacing	~13.5 px center-to-center, to capacity ~33 balls
Speed	~120 px/s (poora loop ~3.7-4 s)
Feed	Ball neck se girta hai (0.12 s drop, ease-in) aur top lane par feed point par snap hota hai

Feed rule: feed point ke aas paas (spacing se kam doori par) koi belt ball ho to gate band. Belt full ho to funnel mein backlog.

9. Brick columns (matching boxes)

```
   belt bottom lane (y=532)
   ==================================
     [o o o]    [o o o]    [o o o]    [o o o]    <- top brick (3 holes)
     [______]   [______]   [______]   [______]   <- 2nd brick (sirf front
face)
     [______]   [______]   [______]   [______]
        .          .          .          .       <- stack screen ke neeche
tak
     col0 x=65  col1 x=120 col2 x=175  col3 x=230
```

- Brick: 50 px wide, top face ~20 px high (3 holes), stack pitch 23 px, radius 6, lego-style: top par 3 hole, neeche wali bricks par sirf front face aur halki top highlight.
- Holes: brick ke color ka darker shade (khali hole), x offset -14, 0, +14, diameter ~12. Bhara hua hole = ball ka color.
- Stacks lambi hain (6 se zyada, screen ke neeche tak). Order level data se aata hai.

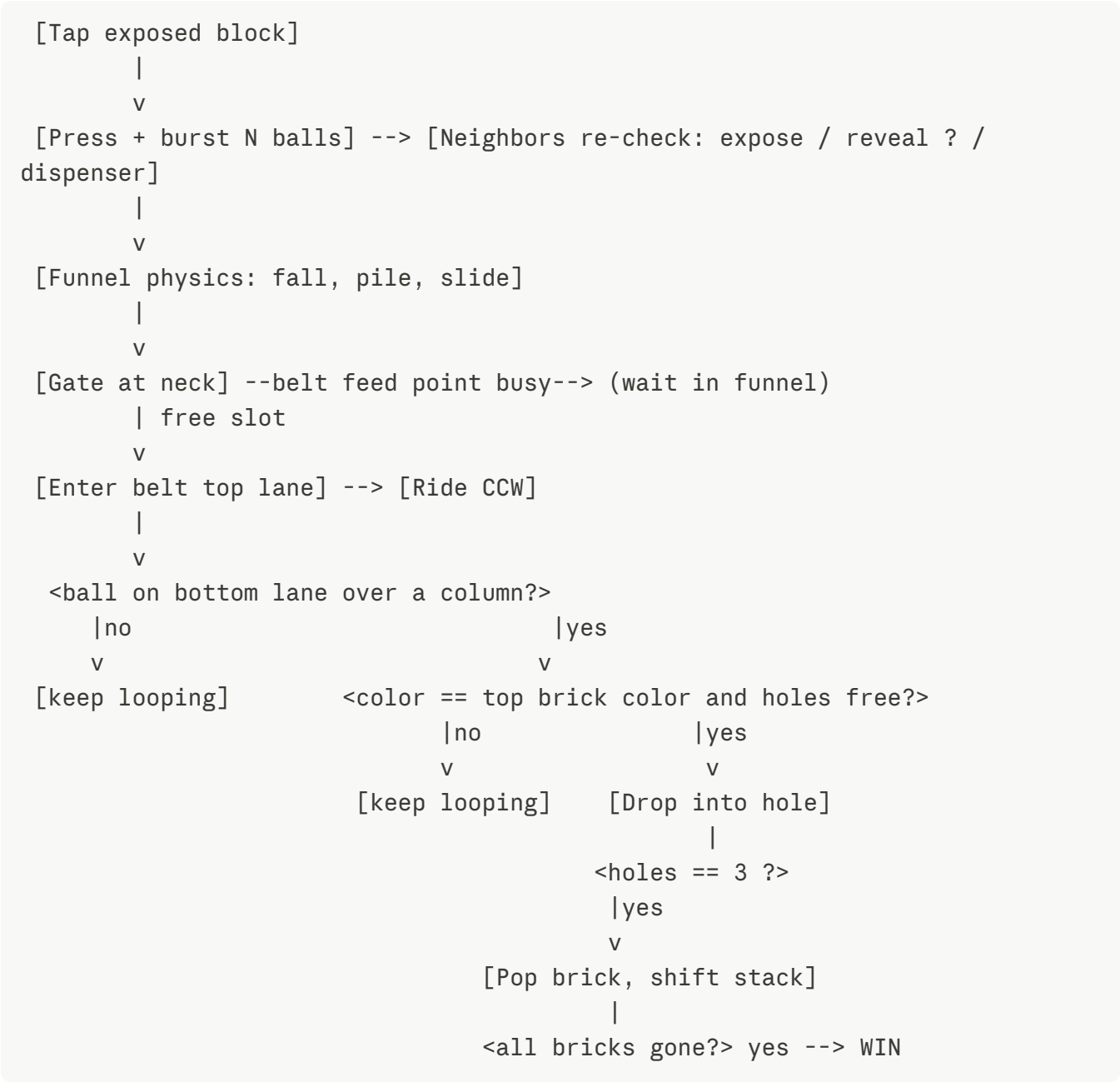
Match logic

1. Belt ka ball bottom lane par kisi column ke x-range [cx-25, cx+25] mein aaye.
2. Agar column ki top brick ka color == ball color AND (filled + inFlight) < 3 to ball belt se nikalta hai.
3. Ball hole ki taraf girta hai (0.18 s, thoda arc), hole bhar jata hai.
4. Match na ho to ball belt par chalta rehta hai aur loop kar ke dobara aata hai.
5. Do columns ka top color same ho (video 0 s par col0 aur col1 dono purple) to ball pehle wale (left, jo pehle aaye) mein jata hai; wo bhar jaye to agle mein.
6. Ek hi frame mein bahut balls na girein: har column ke liye minimum 60 ms gap.

Brick complete hone par

```
0.00 s  3rd ball hole mein pahuncha
0.00-0.10 s  brick scale 1.12 + 10px upar lift (belt ke overlap tak)
0.10 s  3-4 white 4-point stars particles
0.10-0.25 s  brick fade/scale-out
0.10-0.35 s  neeche ki poori stack 23px upar slide (halka overshoot)
0.35 s  nayi top brick par 3 khali holes dikhte hain
```

10. Poora game loop (flow diagram)



11. Data model aur pseudo-code

```
type Color = 'purple'|'pink'|'ice'|'yellow'|'lime'|'cyan'|'orange'|'red';

interface Cell { kind: 'empty'|'block'|'locked'|'dispenser'|'wall';
  color?: Color;            // block ya locked ka hidden color
  counter?: number;         // dispenser
  queue?: Color[];          // dispenser ke naye blocks ka order
}

interface Level {
  grid: Cell[6][6];                // section 5 ka mask
  columns: Color[][];              // 4 stacks, index 0 = top brick
  ballsPerBlock: number;           // default 9
}
```

Exposed check:

```
function isExposed(r, c) {
  if (r === 5) return true;                       // exit row, neeche funnel
  return [[1,0],[-1,0],[0,1],[0,-1]].some(([dr,dc]) =>
    grid[r+dr]?.[c+dc]?.kind === 'empty');        // mask ke andar khali
neighbor
}
```

Tap handler:

```
function onTap(r, c) {
  const b = grid[r][c];
  if (b.kind !== 'block' || !isExposed(r, c)) return;
  playPress(b);                        // 0.1 s squash
  grid[r][c] = EMPTY;
  spawnBalls(center(r, c), b.color, BALLS_PER_BLOCK);
  afterGridChange();
}

function afterGridChange() {
  for each cell:
    if locked and has empty neighbor: reveal();            // flip animation
    if block and becameExposed: popBounce();
    if dispenser and counter > 0 and cell below is empty:  // seedha neeche
      grid[below] = block(dispenser.queue.shift()); counter--; slideIn();
}
```

Belt update (har frame):

```
for (const ball of belt.balls) ball.s = (ball.s + SPEED * dt) % PATH_LEN;
sortBySpacing(); keepMinGap(BALL_SPACING);

for (const ball of belt.balls) {
  if (!onBottomLane(ball.s)) continue;
  const col = columnUnder(ball.x);                    // |x - cx| < 25
  if (!col) continue;
  const brick = col.top();
  if (brick.color === ball.color && brick.filled + brick.inFlight < 3) {
    brick.inFlight++; removeFromBelt(ball); tweenDrop(ball, brick.nextHole(),
0.18,
      () => { brick.inFlight--; brick.filled++; if (brick.filled === 3)
completeBrick(col); });
  }
}

function completeBrick(col) { popAnim(col.top()); col.shiftUp(); checkWin();
}
```

Gate (funnel se belt):

```
const feedFree = !belt.balls.some(b => dist(b.s, FEED_S) < BALL_SPACING);
for (const body of funnelBalls)
  if (body.y > NECK_Y && feedFree) { world.remove(body); belt.insert(FEED_S,
body.color); feedFree = false; }
```

Level solvability rule

Har color ke liye: total balls = (us color ke blocks) x BALLS_PER_BLOCK, aur total bricks = (us color ki bricks) x 3. Dono barabar hone chahiye (BALLS_PER_BLOCK = 9 ho to har block = 3 bricks). Warna level kabhi complete nahi hoga.

12. Animation aur 'juice' ki list

- Block tap: squash + darken, 0.1 s.
- Burst: balls randomly staggered spawn, halki random size jitter nahi, sab same size.
- Expose pop: scale 1 -> 1.15 -> 1, ease-out-back, 0.25 s, texture bumpy.
- Locked reveal: tile y-axis flip ya scale-pop, hidden color nikalta hai.
- Dispenser drop: naya block 0.25 s mein neeche slide, counter number bounce.
- Ball feed: neck se belt par gir kar chhota squash.
- Ball into hole: 0.18 s arc, hole mein chhota scale-pop.
- Brick complete: lift + stars + fade, stack slide up with overshoot.
- Haptics/sound (optional): tap pop, ball clink, brick ding.
- Ball shading: glossy circle, halka highlight spot, thin darker outline (white balls par grey outline).

13. Jo video mein NAHI dikha (assumptions, flag karke build karo)

- Win: saari bricks complete ho jayen (ya sab blocks clear aur belt khali).
- Lose: belt full ho aur funnel mein balls wait kar rahe hon aur belt par koi ball kisi top brick se match na kare, 2-3 sec ke baad fail.
- Power-ups (magnet, shuffle, gumball) video mein use nahi hue. Guess: magnet = ek color ke balls ko belt se khinch kar brick mein, shuffle = grid ke blocks ke colors mix, gumball = random bonus. Pehle version mein buttons rakho aur effect baad mein.
- Coins, Shop, Settings, Restart ki functionality.
- Exact ball counts per block aur belt ki exact speed: constants tunable rakho.

- Dispenser counter 0 hone par kya hota hai (assume: gayab ho jata hai aur cell khali).

14. AI Studio ke liye ready prompt (copy-paste)

```
Build a single-file HTML5 canvas mobile game called 'Marble Sort' (portrait,
reference resolution 296x640, scale everything by width/296). Use Matter.js
(CDN) only for the funnel ball physics; everything else is custom.

LAYOUT (y from top): HUD row at y36-68 (shop button, coin pill '2,450',
'Level 35' pill, settings gear), restart button under gear. Power-up row
y118-174 with 3 green square buttons (magnet, shuffle, gumball) each with a
red badge '1'. Glass container x8-280, y182-432 (fill #93D8E4, border #427E8E
5px, top corner radius 32). Inside: stepped pyramid block grid, 6 cols x 6
rows, pitch 35 x 33, cell 30x30 radius 7. Below y432: funnel with walls
sloping 22 degrees inward to a 38px neck at y487 (fill #C7D2EB). Below the
neck: a stadium-shaped conveyor belt x28-267, y498-542 with a center divider
bar. Below the belt: 4 brick columns at x=65,120,175,230.

GRID (r0..r5, c0..c5, . = wall, ? = locked, D3 = dispenser counter 3):
r0: . . ? ? . .
r1: . ? red purple ? .
r2: ? yellow cyan purple ice ?
r3: D3 ? purple lime ? lime
r4: purple orange yellow D3 pink yellow
r5: . lime ice pink purple .   (exit row)
Give '?' tiles a hidden color each. Dispensers have a queue of colors.

RULES:
1. A block is tappable only if one of its 4 neighbors is an empty in-mask
cell, or it is in row r5. Tappable blocks show a bumpy 3x3 dot texture and do
a scale pop (1->1.15->1, 0.25s) when they become exposed. No gravity on
blocks.
2. Tap: squash 0.1s, remove block, burst 9 balls (radius 7) with random
velocity 60-140 px/s, staggered 0-60ms. Balls do NOT collide with blocks,
only with container walls, funnel walls and each other. Render balls above
blocks.
3. Funnel physics: gravity 1000 px/s^2, restitution 0.3, friction 0.02. Balls
pile up and slide to the neck. An invisible gate at the neck lets a ball
through only when the belt feed point (x=147, top lane) has a free slot (no
belt ball within 13.5px), otherwise balls wait in the funnel.
4. Belt: path-following loop, counter-clockwise: top lane y=508 moving left,
left U-turn (center (57,520), r=12), bottom lane y=532 moving right, right U-
```

```
turn (center (243,520)). Length ~447px, ball radius 6, spacing 13.5px, speed
120 px/s, capacity ~33. Draw dark pill slots along the lanes.
5. Brick columns: each column is a stack (index 0 = top). Top brick shows 3
holes (darker shade of its color). When a belt ball is on the bottom lane
within 25px of a column center and its color equals that column's top brick
color and filled+inFlight < 3, the ball leaves the belt and drops into the
next hole (0.18s arc). Min 60ms between drops per column. If it does not
match, it keeps looping.
6. When a brick has 3 filled holes: lift 10px + scale 1.12, white star
particles, fade out, the rest of the stack slides up 23px with slight
overshoot, next brick becomes top.
7. Locked '?' tiles reveal (flip animation) when any orthogonal neighbor
becomes empty. Dispensers: when the cell directly below is empty and
counter>0, slide a new block (from its queue) into that cell and decrement
the counter.
8. Win when all bricks are complete. Lose when the belt is full, funnel has
waiting balls, and no belt ball matches any top brick for 3 seconds. Show
simple win/lose overlays with a restart button.
9. Level data must satisfy: for every color, (number of blocks of that color
x 9) == (number of bricks of that color x 3). Provide a level generator or a
hand-made Level 35 that satisfies this.

VISUAL STYLE: candy/hyper-casual, flat teal background #56A9B4, glossy balls
with thin outline, rounded blocks with a 3px darker bevel, white sparkle
stars on brick complete. Colors: purple #7B3FE4, pink #FF8FD6, ice #E6F4FA,
yellow #FFD21F, lime #A8F028, cyan #27C4F4, orange #FF8A1F, red #F23B5C.

Expose all tunables as constants at the top (BALLS_PER_BLOCK, BELT_SPEED,
BALL_SPACING, GRAVITY, NECK_WIDTH, BRICK_HOLES).
```

15. Build order (AI Studio ko step by step dena)

1. Layout aur borders: sirf shapes, colors, HUD (kuch chalta nahi).
2. Block grid + exposed rule + tap + burst balls (funnel physics ke bina, balls neeche gir kar gayab).
3. Funnel walls + Matter.js physics + gate at neck.
4. Belt path-following + feed.
5. Brick columns + match + complete + stack shift.
6. Locked '?' reveal + dispenser.
7. Win/lose + restart + level data.

8. Juice: animations, particles, sound, power-up stubs.